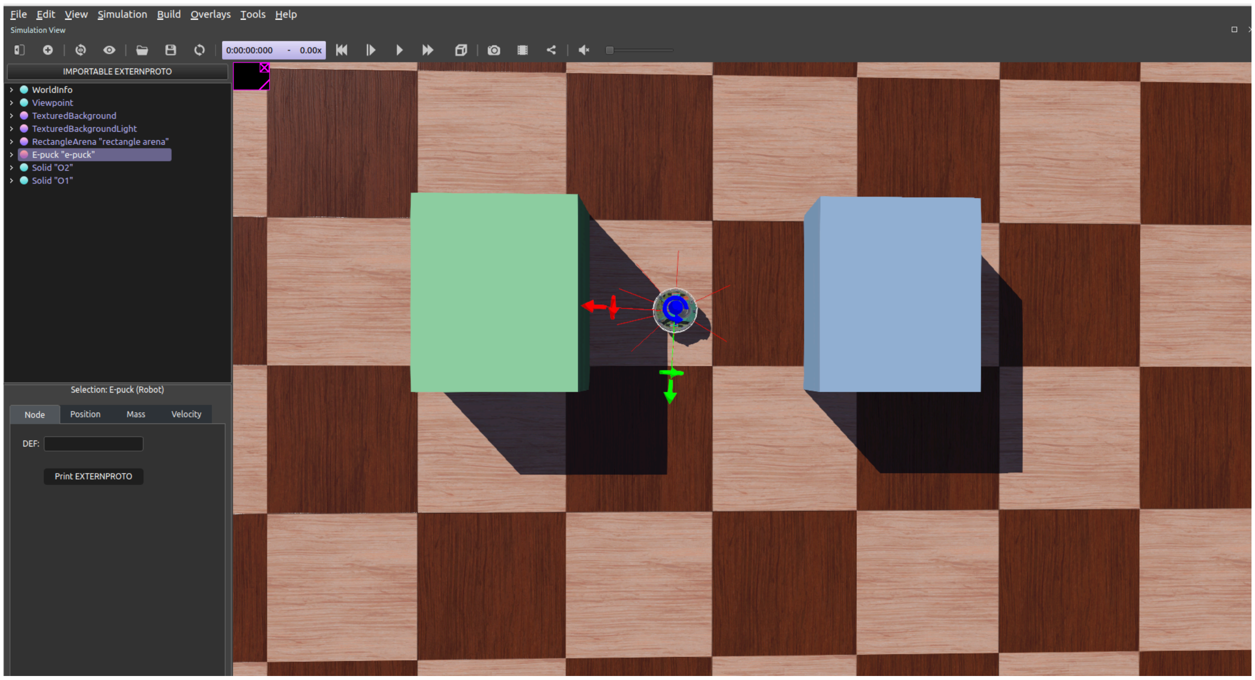
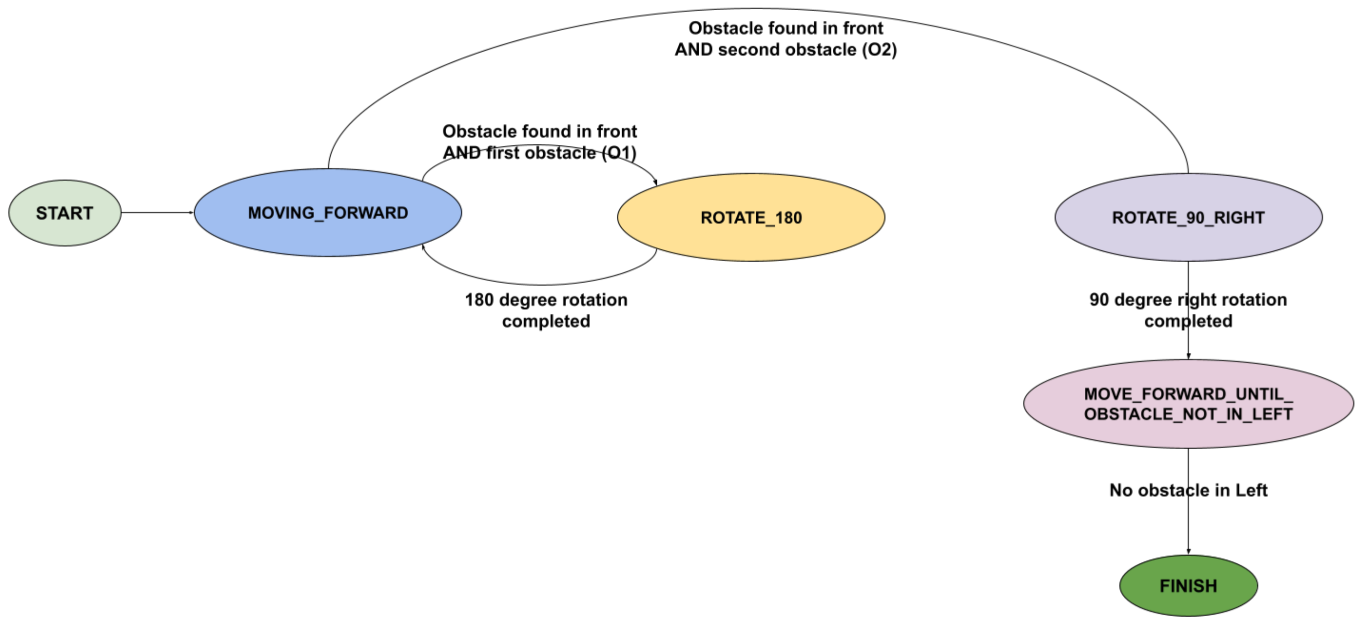
Assignment-3: **Basic Reactive Behaviors**

In this assignment, a basic reactive control with pre-defined behaviour and a finite state machine is implemented.

The world contains two obstacles (**O1 box = Green**, **O2 box = Blue**) and an e-puck robot inbetween As shown in the image below:



Finite state machine (FSM) diagram



Mechanisms implemented in States

MOVING_FORWARD
A velocity of PI (3.14) radian is set for both left and right motors equally to move forward.
This state persists until an obstacle is detected in the front proximity sensors (7 and 0).

ROTATE_180
To rotate 180 degrees, a velocity of -PI is set to the left motor, and a velocity of +PI is set to the right motor, for a timespan of ~1.5 seconds. Once the rotation is completed, state is assigned as MOVING_FORWARD.

ROTATION_90_RIGHT
Once the second obstacle is found, the robot is rotated 90 degrees to the right.
To rotate 180 degrees, a velocity of -PI is set to the left motor, and a velocity of +PI is set to the right motor, for a timespan of ~0.75 seconds.

MOVE_FORWARD_UNTIL_OBSTACLE_NOT_IN_LEFT
Once the 90 degree right rotation is completed, the robot is assigned with MOVE_FORWARD_UNTIL_OBSTACLE_NOT_IN_LEFT state, where the left proximity sensor is observed until the proximity value is less than a threshold.

FINISH
When there is not obstacle in the left side, the robot is stopped by assigning 0 velocity to both wheel motors.